In [55]:
```
import numpy as np
from numpy.random import randn
from scipy.linalg import sqrtm, inv, norm
import matplotlib.pyplot as plt
import matplotlib as mpl
from scipy.stats import chi2
```

In [63]:
```
def epsilon_and_nu(P: float, Sigma: np.ndarray):
    nu = 1/ (2 * np.pi * norm(sqrtm(Sigma)))
    epsilon = (1-P) * nu

    return epsilon, nu

# If inside the ellipse
def in_error_ellipse(x: np.ndarray, mu: np.ndarray, Sigma: np.ndarray):
    P = 0.95
    epsilon, nu = epsilon_and_nu(P, Sigma)

    # gaddamn why is this so complicated
    return bool(((x-mu).reshape((1,2)) @ inv(Sigma) @ (x-mu) <= -2*np.log(epsilon/nu))[0

def plot_gaussian(mu, Sigma, N):
    x_arr = randn(N,2) @ sqrtm(Sigma) + mu

    x_in = np.array([x for x in x_arr if in_error_ellipse(x, mu, Sigma)])
    x_out = np.array([x for x in x_arr if not in_error_ellipse(x, mu, Sigma)])

    e, _ = epsilon_and_nu(0.95, Sigma)
    print(f"Percentage of points in ellipse {len(x_in) / N * 100}")
    plt.scatter(x_in[:,0], x_in[:,1], marker='.')
    plt.scatter(x_out[:,0], x_out[:,1], marker='.')
    plt.title("95% Confidence Error Ellipse")
    plt.show()

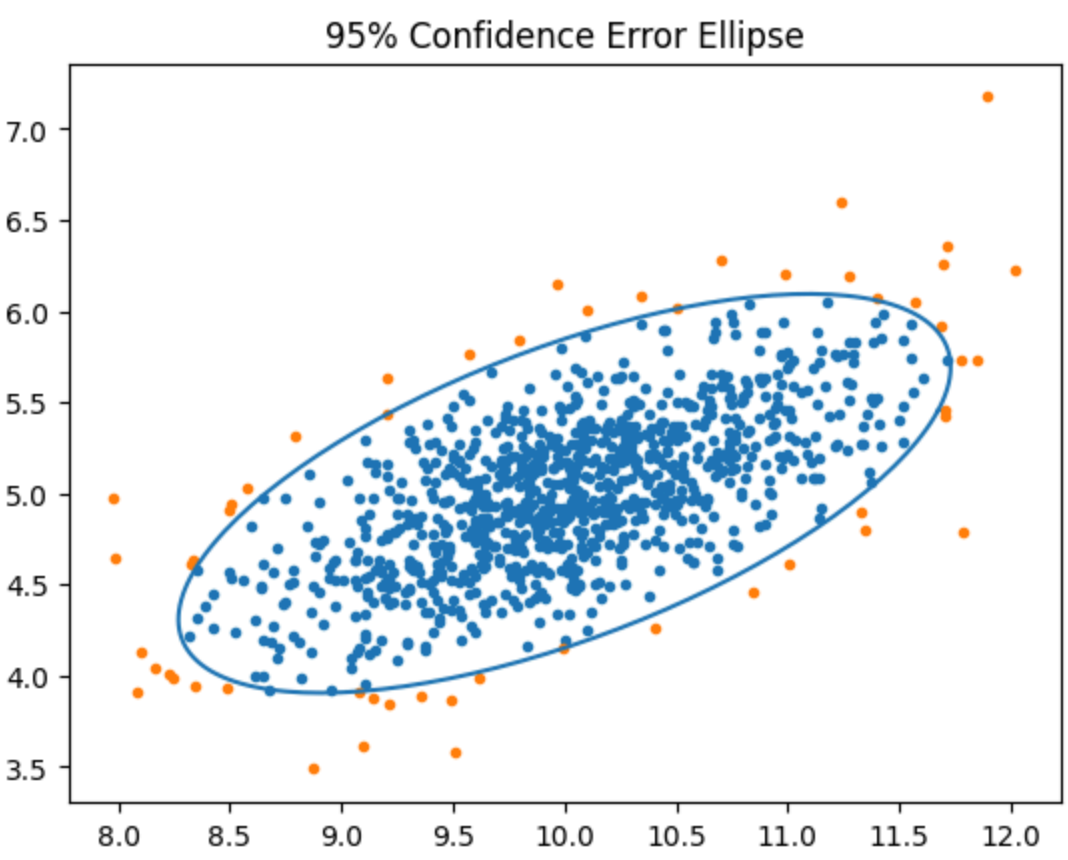
def get_ellipse_points(mu: np.ndarray, Sigma: np.ndarray, level=0.95, n=200):
    t = np.linspace(0, 2*np.pi, n)
    circle = np.vstack((np.cos(t), np.sin(t)))
    r = np.sqrt(chi2.ppf(level, df=2))
    ellipse = sqrtm(Sigma) @ (r * circle)
    return ellipse + mu.reshape(2,1)
```

In [64]:
```
mu_1 = np.array([10,5])
Sigma_1 = np.array([
    [0.5, 0.2],[0.2, 0.2]
])



plt.plot(*get_ellipse_points(mu_1, Sigma_1))
plot_gaussian(mu_1, Sigma_1, 1000)
```

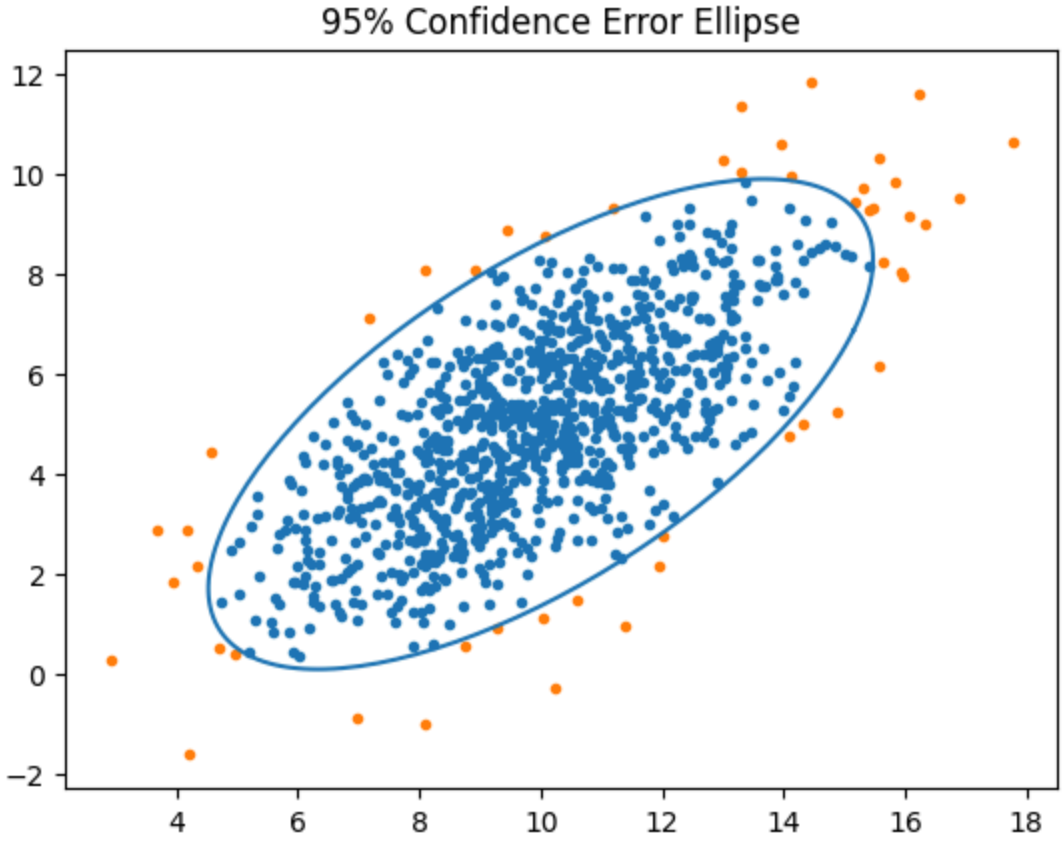
```
Percentage of points in ellipse 94.6
```



In [66]:
```
mu_2 = np.array([10,5])
Sigma_2 = np.array([
    [5, 3],[3, 4]
])

plt.plot(*get_ellipse_points(mu_2, Sigma_2))
plot_gaussian(mu_2, Sigma_2, 1000)
```

```
Percentage of points in ellipse 95.1
```



In [67]:
```
mu_3 = np.array([3,5])
Sigma_3 = np.array([
    [1.4, 3],[3, 8]
])

plt.plot(*get_ellipse_points(mu_3, Sigma_3))
plot_gaussian(mu_3, Sigma_3, 1000)
```

```
Percentage of points in ellipse 94.6
```

